**4** **(a)** Define *electromotive force* of a cell, and explain its difference from the *terminal potential difference* across the terminals of the cell.

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[2]

**The electromotive force of a cell is the energy converted from other forms of energy into electrical energy per unit charge driven round a complete circuit.** **B1**

**Terminal potential difference across the battery however is the e.m.f of the cell less the loss volts across the internal resistance of the cell.** **B1**

**(b)** The variation with current $I$ of terminal potential difference $V$ across a battery with e.m.f. $E$ and internal resistance $r$ is shown in Fig. 4.1.

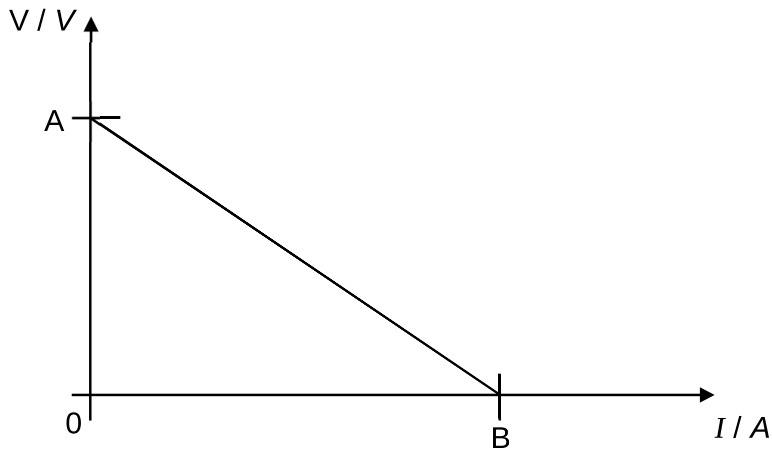



**Fig. 4.1**

In Fig. 4.1, state and explain the significance of the values of

**(i)** the vertical intercept, A;

[2]

**The vertical intercept is the value of the e.m.f. of the battery.** **M1**

**This is the potential difference across the battery when there is no current flowing through it hence zero potential difference across the internal resistance.** **A1**

**(ii)** the horizontal intercept, B.

[2]

**The horizontal intercept is the value of the maximum current that will flow through the battery when there is no (external) resistance applied across it. (i.e. short circuit current)** **M1** **A1**

**(c)** In an attempt to obtain the graph of Fig. 4.1 for the battery, a student sets up a circuit as shown in Fig. 4.2.

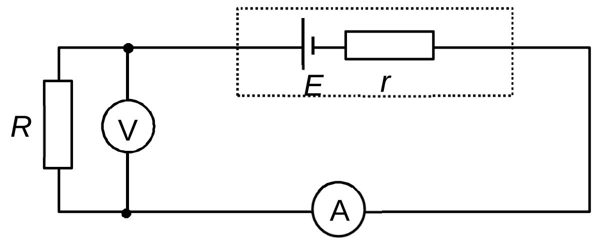



**Fig. 4.2**

State and explain why the circuit shown in Fig. 4.2 is inappropriate for obtaining data for the full range of values in Fig. 4.1.

[4]

**The resistor R used is a fixed resistor.** **B1**

**This would not allow the current to vary in Fig. 4.1, but the graph in Fig. 4.1 shows that there is a varying current in the circuit.** Thus it is not possible to have the variation of potential difference plotted while current is varying. **B1**

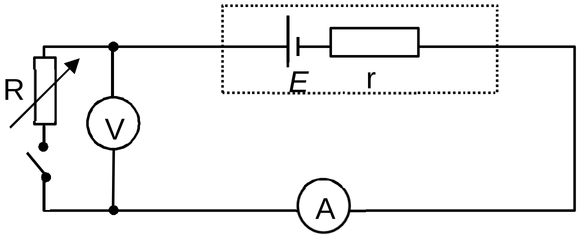
In addition, the **voltmeter will not be able to measure the e.m.f. of the battery as well as the current at lower values / voltage at higher values even when the fixed resistor has been replaced by a rheostat/variable resistor .** **B1**

**This is because** the rheostat cannot be adjusted to infinite resistance and thus **there would always be some current flowing through the circuit.** **B1**

**(d)** In the space below draw a circuit diagram, using the appropriate components, from which the full range of values in Fig. 4.1 may be obtained.

[2]

**Solution:**




- **Replace the fixed resistor with a rheostat.** **B1**
- **Add in a switch in series with the rheostat.** **B1**